

6. SYSTEM DESIGN

6.1 Use Case Diagram

This section presents the Use Case Diagram of MediSync. It shows who uses the system, the major functions each user can perform, and how each user connects to those functions. The diagram covers only the major functions listed in the Functional Requirements in Chapter 3. Each use case carries its requirement ID from Table 3.1, so it can be traced back to the requirement it comes from. Figure 6.1 shows the diagram above.

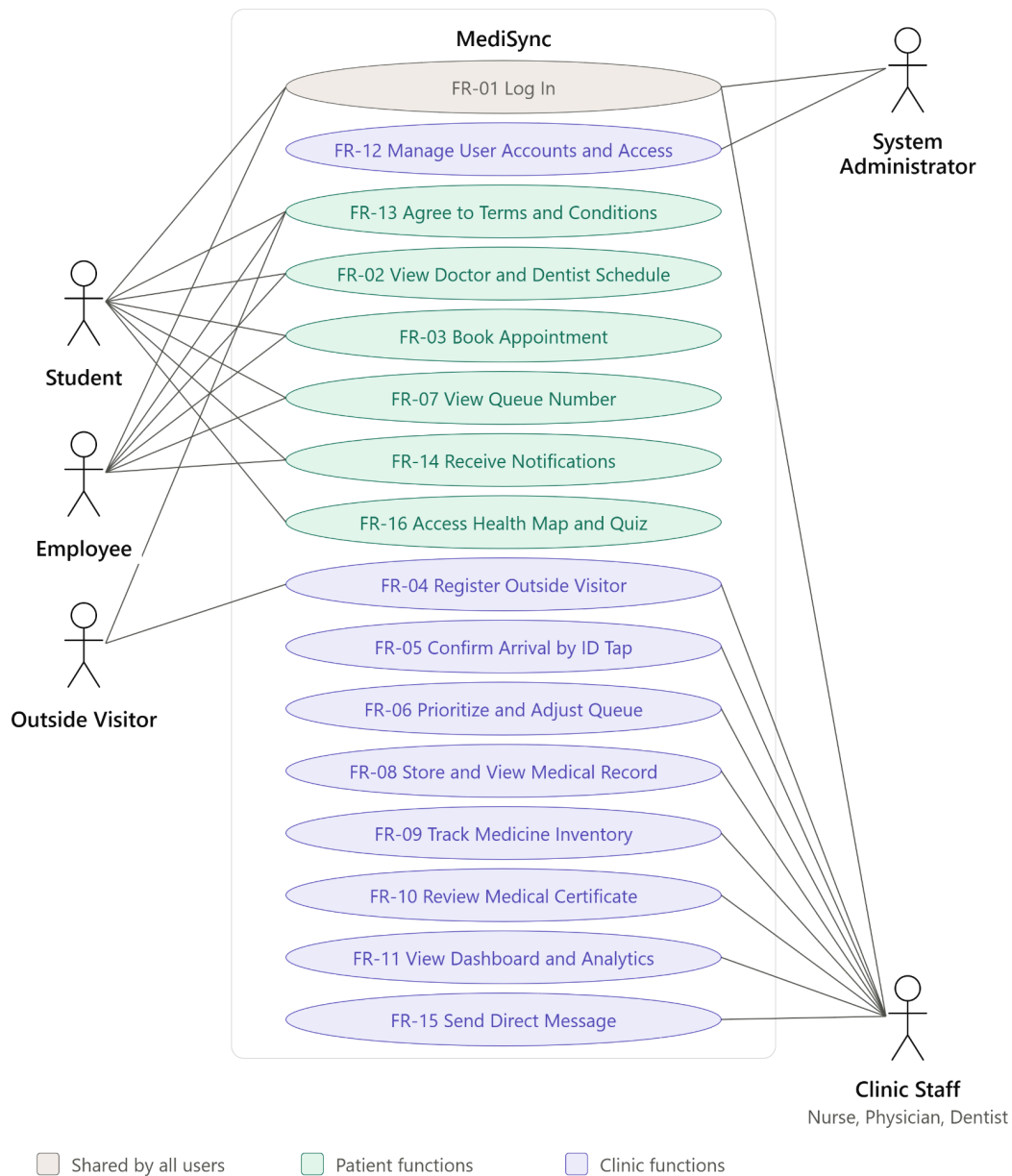


Fig. 6.1.1 Use Case Diagram of MediSync

The diagram has five actors. The Student, the Employee, and the Outside Visitor are the patients who receive care from the clinic. The Clinic Staff actor stands for the nurses, physicians, and dentists, since the Functional Requirements describe their functions under clinic staff as one group. The System Administrator is the actor who controls who can use the system and what each person can access.

Log In is shared by the Student, Employee, Clinic Staff, and System Administrator. The Outside Visitor does not log in because clinic staff register visitors manually at the time of the visit, so the visitor connects only to Agree to Terms and Conditions and Register Outside Visitor. The Student and Employee can agree to the terms, view the doctor and dentist schedule, book an appointment, view their queue number, and receive notifications. Only the Student can access the health map and quiz.

The Clinic Staff handle the daily clinic work. They register outside visitors, confirm a patient's arrival by ID tap, and review the suggested priority level to adjust the queue. They also store and view medical records, track medicine inventory, review the generated medical certificate, view the dashboard and analytics, and send direct messages to students or parents. The System Administrator is connected to Manage User Accounts and Access, which gives each type of user the right level of access.

The colors in the diagram group the use cases by who mainly uses them. Gray is shared by all users, teal is for patient functions, and purple is for clinic functions.

6.2 Context Diagram

This section presents the Context Diagram of MediSync. It shows the whole system as one process and the outside parties that exchange data with it. The inside of the system is not shown here. Figure 6.2 shows the diagram above.

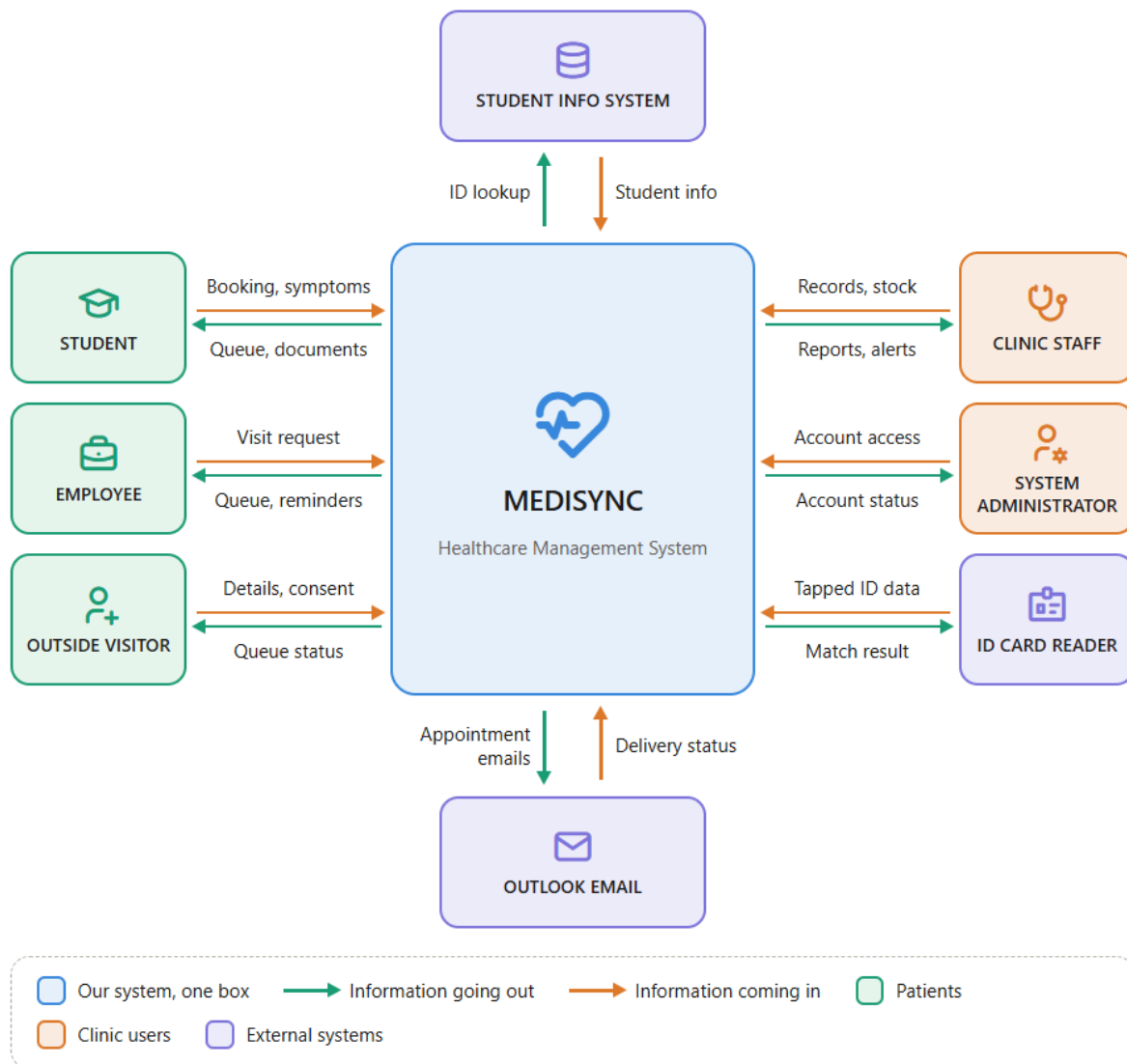


Fig. 6.2.1 Context Diagram of MediSync

The diagram has eight external entities. Five are people who use the system, which are the Student, the Employee, the Outside Visitor, the Clinic Staff, and the System Administrator. Clinic Staff stands for the nurses, physicians, and dentists, the same group used in the Use Case Diagram. The other three are outside systems that MediSync works with. These are the School SIS, which is the school's existing Student Information System, the ID Card Reader used for tap in check in, and Outlook Email, which is the university email service.

Data entering the system comes from each entity. Students send their login details, appointment bookings, and symptoms. Employees send their login details and visit requests. Outside Visitors give their basic details and consent so clinic staff can register them. Clinic Staff enter patient records, vital signs, consultation notes, and medicine stock data. The System Administrator sends account and

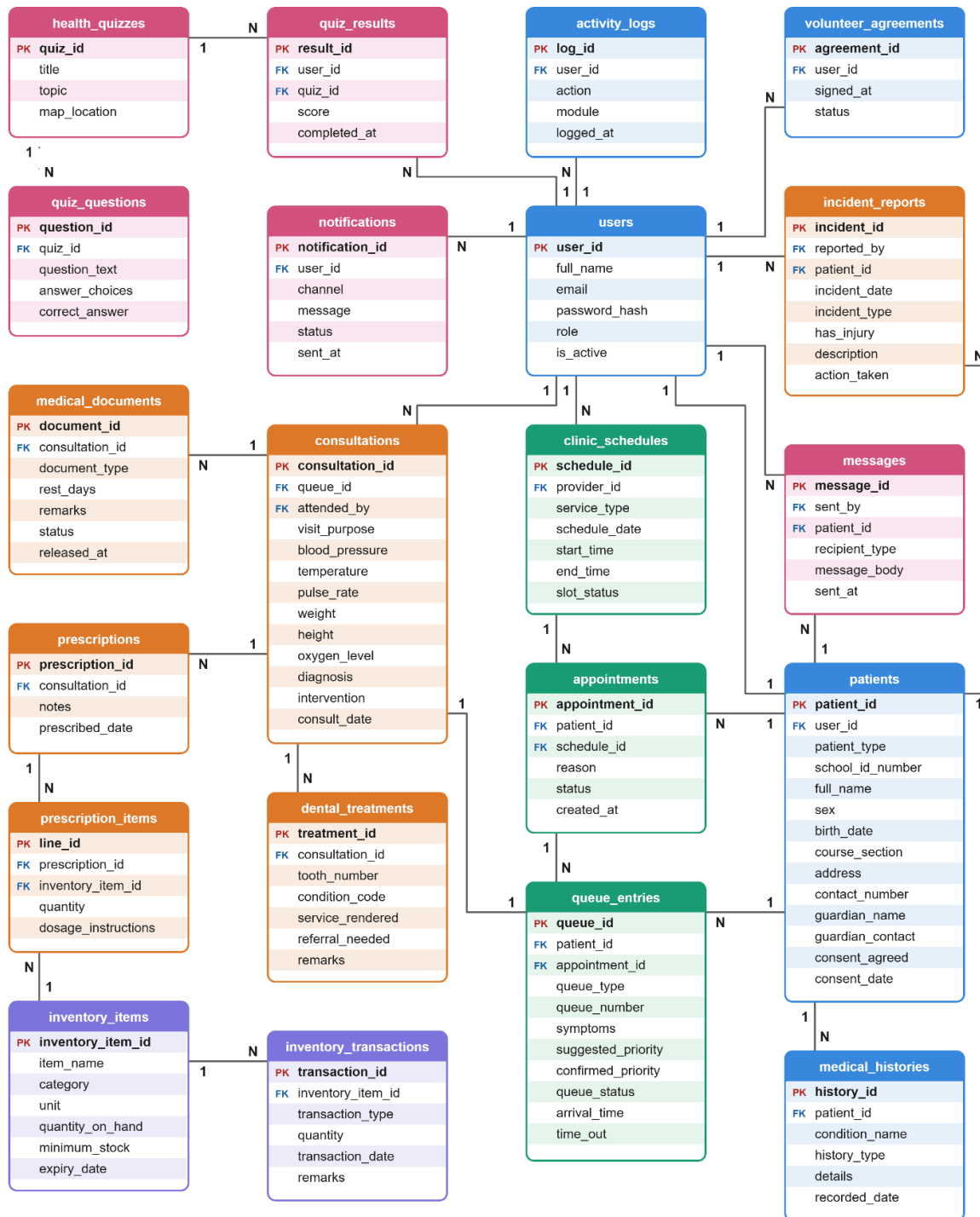


access settings. The School SIS supplies student basic information, the ID Card Reader sends the tapped ID data, and Outlook Email returns the delivery status of sent messages.

Data produced by the system goes back to the same entities. Students and Employees receive their queue number, reminders, and released documents such as prescriptions and medical certificates. Outside Visitors receive their queue status. Clinic Staff receive reports, dashboard analytics, and low stock alerts. The System Administrator receives the status of user accounts. MediSync sends a student ID lookup to the School SIS, a match result to the ID Card Reader, and appointment emails and notices to Outlook Email.

6.3 Database Schema

This section presents the proposed database design of MediSync. It shows the main tables, their important attributes, the primary keys, the foreign keys, and how the tables connect to each other. The design follows the centralized database described in Chapter 3, where records from different modules are linked to the same student and the same visit. Figure 6.3 shows the schema diagram.



PK Primary key FK Foreign key 1 One N Many
 Accounts and access Scheduling and queue Consultation and records Inventory Communication and quizzes

Fig. 6.3.1 Database Schema of MediSync

The schema has 21 tables. They are grouped into five modules, which are accounts and access, scheduling and queue, consultation and records, inventory, and communication and quizzes. Each table lists its primary key, marked PK, its foreign keys, marked FK, and its key attributes. A line between two tables means the foreign key in one table points to the primary key of the other. A relationship is either one to one or one to many, and Table 6.1 lists each one.

The accounts and access module keeps the people and their permissions. The users table holds the login accounts of students, employees, clinic staff, administrators, and student volunteers. The role column decides what each person can open, which supports the role based access requirement. The patients table is separate from users because outside visitors have no account. A student or employee patient is linked to an account through user_id, while the user_id of a visitor stays empty because clinic staff register visitors manually. The patients table also holds the consent agreement, so the consent of a visitor is saved even without an account. The medical_histories table stores the personal and family health conditions listed in the school health examination card, and each record belongs to one patient. The volunteer_agreements table keeps the signed agreement of each student volunteer, and the activity_logs table records what each user does in the system for the administrator to review.

The scheduling and queue module follows the clinic flow from booking to arrival. The clinic_schedules table holds the doctor and dentist schedules that students see, and the appointments table links a patient to one schedule slot. The queue_entries table holds the symptoms, the suggested priority, the confirmed priority that clinic staff review, and the queue type, which separates the priority queue from the separate queue for busy days. Its appointment_id stays empty for a patient who arrives without an appointment, which matches the walk in rule. The time in and time out of each visit are also kept here, which supports the daily clinic log sheet.

The consultation and records module holds what happens during the visit. Each consultation is linked to its queue entry, so the visit history of a patient can be traced through the queue and the patient. The consultations table stores the vital signs, the diagnosis, the intervention, and the staff member who attended the patient. For dental visits, the dental_treatments table records the tooth number, the condition, and the service rendered, and it marks when a referral is needed. A consultation can also lead to prescriptions and documents. The prescription_items table lists each medicine in a prescription and points to the inventory, while the medical_documents table stores the medical certificate, excuse letter, medical pass, or go home form based on its document_type. The incident_reports table keeps the incident forms filled out by clinic staff, and each report can be linked to the patient involved.

The inventory module uses two tables. The inventory_items table keeps the stock count and the minimum stock level used for low stock alerts. Every received, dispensed, or adjusted quantity is

recorded in inventory_transactions, so the stock history stays traceable.

The communication and quizzes module covers messages and the health education map. The notifications table stores the in app, Outlook email, and SMS notices for each user, and the messages table stores direct messages from clinic staff about a patient. The health_quizzes table keeps each quiz and its place on the campus map, the quiz_questions table keeps the questions and answer choices, and the quiz_results table keeps the score of each student.

6.4 Screen Layout / User Interface Design

Guide:

Present the proposed design of the major system screens.

Include only the important screens necessary to demonstrate the main system features.

For each screen, provide:

- Screen name
- Screen name
- Brief description of the main components

Screens may be designed using an appropriate prototyping or design tool.

APPENDICES